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TRAVAUX PRATIQUES ENCADRÉS

UE : Supervision, Analyse Forensique et Gestion des incidents

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Matricule	
Niveau	Master II

Année Académique : 2025-2026

SUITE DU TRAVAIL PRATIQUE DU CHAPITRE II

Ex. 2.1 — Analyse d'un fichier access.log.


```
[
  {
    "line": 1,
    "ip": "192.168.1.25",
    "user": "admin",
    "datetime": "12/Apr/2026:03:14:05 +0000",
    "request": {
      "method": "GET",
      "uri": "/index.php",
      "uri_decoded": "/index.php",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 200,
      "size": 4523
    },
    "referer": "https://lizbiz.cm/",
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0
Safari/537.36",
    "analysis": {
      "sql_injection": false,
      "threat_level": "LÉGITIME",
      "threat_detail": "Trafic normal"
    }
  },
  {
    "line": 2,
    "ip": "203.0.113.42",
```

```
    "user": null,
    "datetime": "12/Apr/2026:03:14:10 +0000",
    "request": {
      "method": "POST",
      "uri": "/wp-login.php",
      "uri_decoded": "/wp-login.php",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 401,
      "size": 1234
    },
    "referer": null,
    "user_agent": "sqlmap/1.7#stable (https://sqlmap.org)",
    "analysis": {
      "sql_injection": false,
      "threat_level": "CRITIQUE",
      "threat_detail": "SQLi / sqlmap"
    }
  },
  {
    "line": 3,
    "ip": "203.0.113.42",
    "user": null,
    "datetime": "12/Apr/2026:03:14:12 +0000",
    "request": {
      "method": "POST",
      "uri": "/wp-login.php",
      "uri_decoded": "/wp-login.php",
      "protocol": "HTTP/1.1"
    },
```

```
  "response": {
    "status": 401,
    "size": 1234
  },
  "referer": null,
  "user_agent": "sqlmap/1.7#stable (https://sqlmap.org)",
  "analysis": {
    "sql_injection": false,
    "threat_level": "CRITIQUE",
    "threat_detail": "SQLi / sqlmap"
  }
},
{
  "line": 4,
  "ip": "203.0.113.42",
  "user": null,
  "datetime": "12/Apr/2026:03:14:14 +0000",
  "request": {
    "method": "POST",
    "uri": "/wp-login.php",
    "uri_decoded": "/wp-login.php",
    "protocol": "HTTP/1.1"
  },
  "response": {
    "status": 401,
    "size": 1234
  },
  "referer": null,
  "user_agent": "sqlmap/1.7#stable (https://sqlmap.org)",
  "analysis": {
    "sql_injection": false,
```

```
      "threat_level": "CRITIQUE",
      "threat_detail": "SQLi / sqlmap"
    }
  },
  {
    "line": 5,
    "ip": "192.168.1.25",
    "user": "admin",
    "datetime": "12/Apr/2026:03:14:22 +0000",
    "request": {
      "method": "GET",
      "uri": "/products.php?id=15",
      "uri_decoded": "/products.php?id=15",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 200,
      "size": 2341
    },
    "referer": "https://lizbiz.cm/",
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36",
    "analysis": {
      "sql_injection": false,
      "threat_level": "LÉGITIME",
      "threat_detail": "Trafic normal"
    }
  },
  {
    "line": 6,
    "ip": "203.0.113.42",
    "user": null,
```

```
    "datetime": "12/Apr/2026:03:14:25 +0000",
    "request": {
      "method": "GET",
      "uri":
"/products.php?id=15%20UNION%20SELECT%201,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20%20FROM%20information_schema.tables--",
      "uri_decoded": "/products.php?id=15 UNION SELECT 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20 FROM information_schema.tables--",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 500,
      "size": 567
    },
    "referer": null,
    "user_agent": "sqlmap/1.7#stable (https://sqlmap.org)",
    "analysis": {
      "sql_injection": true,
      "threat_level": "CRITIQUE",
      "threat_detail": "SQLi / sqlmap"
    }
  },
  {
    "line": 7,
    "ip": "203.0.113.42",
    "user": null,
    "datetime": "12/Apr/2026:03:14:30 +0000",
    "request": {
      "method": "GET",
      "uri": "/products.php?id=15%27%20AND%20SLEEP(5)--",
      "uri_decoded": "/products.php?id=15' AND SLEEP(5)--",
      "protocol": "HTTP/1.1"
```

```
    },
    "response": {
      "status": 500,
      "size": 567
    },
    "referer": null,
    "user_agent": "sqlmap/1.7#stable (https://sqlmap.org)",
    "analysis": {
      "sql_injection": true,
      "threat_level": "CRITIQUE",
      "threat_detail": "SQLi / sqlmap"
    }
  },
  {
    "line": 8,
    "ip": "10.0.1.5",
    "user": null,
    "datetime": "12/Apr/2026:03:15:01 +0000",
    "request": {
      "method": "GET",
      "uri": "/healthcheck",
      "uri_decoded": "/healthcheck",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 200,
      "size": 15
    },
    "referer": null,
    "user_agent": "InternalMonitoring/1.0",
    "analysis": {
```

```
    "sql_injection": false,
    "threat_level": "NORMAL",
    "threat_detail": "Monitoring interne"
  }
},
{
  "line": 9,
  "ip": "203.0.113.42",
  "user": null,
  "datetime": "12/Apr/2026:03:15:05 +0000",
  "request": {
    "method": "GET",
    "uri": "/admin/config.php.bak",
    "uri_decoded": "/admin/config.php.bak",
    "protocol": "HTTP/1.1"
  },
  "response": {
    "status": 404,
    "size": 345
  },
  "referer": null,
  "user_agent": "Mozilla/5.0 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)",
  "analysis": {
    "sql_injection": false,
    "threat_level": "MOYEN",
    "threat_detail": "Reconnaissance"
  }
},
{
  "line": 10,
  "ip": "5.188.62.18",
```

```
    "user": null,
    "datetime": "12/Apr/2026:03:16:11 +0000",
    "request": {
      "method": "POST",
      "uri": "/api/v1/login",
      "uri_decoded": "/api/v1/login",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 401,
      "size": 256
    },
    "referer": null,
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36",
    "analysis": {
      "sql_injection": false,
      "threat_level": "ÉLEVÉ",
      "threat_detail": "Brute Force"
    }
  },
  {
    "line": 11,
    "ip": "5.188.62.18",
    "user": null,
    "datetime": "12/Apr/2026:03:16:15 +0000",
    "request": {
      "method": "POST",
      "uri": "/api/v1/login",
      "uri_decoded": "/api/v1/login",
      "protocol": "HTTP/1.1"
    },
```

```
    "response": {
      "status": 401,
      "size": 256
    },
    "referer": null,
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36",
    "analysis": {
      "sql_injection": false,
      "threat_level": "ÉLEVÉ",
      "threat_detail": "Brute Force"
    }
  },
  {
    "line": 12,
    "ip": "5.188.62.18",
    "user": null,
    "datetime": "12/Apr/2026:03:16:19 +0000",
    "request": {
      "method": "POST",
      "uri": "/api/v1/login",
      "uri_decoded": "/api/v1/login",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 401,
      "size": 256
    },
    "referer": null,
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36",
    "analysis": {
```

```
      "sql_injection": false,
      "threat_level": "ÉLEVÉ",
      "threat_detail": "Brute Force"
    }
  },
  {
    "line": 13,
    "ip": "5.188.62.18",
    "user": null,
    "datetime": "12/Apr/2026:03:16:23 +0000",
    "request": {
      "method": "POST",
      "uri": "/api/v1/login",
      "uri_decoded": "/api/v1/login",
      "protocol": "HTTP/1.1"
    },
    "response": {
      "status": 401,
      "size": 256
    },
    "referer": null,
    "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36",
    "analysis": {
      "sql_injection": false,
      "threat_level": "ÉLEVÉ",
      "threat_detail": "Brute Force"
    }
  }
]
```

A — IPs générant des codes 401

```
grep '"status": 401' access_parsed.json | grep -oP '"ip": "\K[^"]+'
# Ou directement sur le log brut :
awk '$9 == 401 {print $1}' access.log | sort | uniq -c | sort -rn

# Résultat :
      4  5.188.62.18
      3  203.0.113.42
```

B — URIs ciblées par tentatives SQLi

```
grep -A2 '"sql_injection": true' access_parsed.json | grep uri_decoded
# Ou sur le log brut :
grep -iE "union|select|sleep|%27|%20" access.log | awk '{print $7}' | sort -u

# Résultat :
/products.php?id=15%20UNION%20SELECT%201,2,3...20%20FROM%20information_schema.tables--
/products.php?id=15%27%20AND%20SLEEP(5)--
```

C — User-Agents suspects

```
grep -E "sqlmap|nikto|nmap|curl|python|masscan" access.log | awk -F'"' '{print $6}' | sort
| uniq -c

# Résultat :
      5  sqlmap/1.7#stable (https://sqlmap.org)
```